

[Aim: 100/100 in Maths]

आरंभ

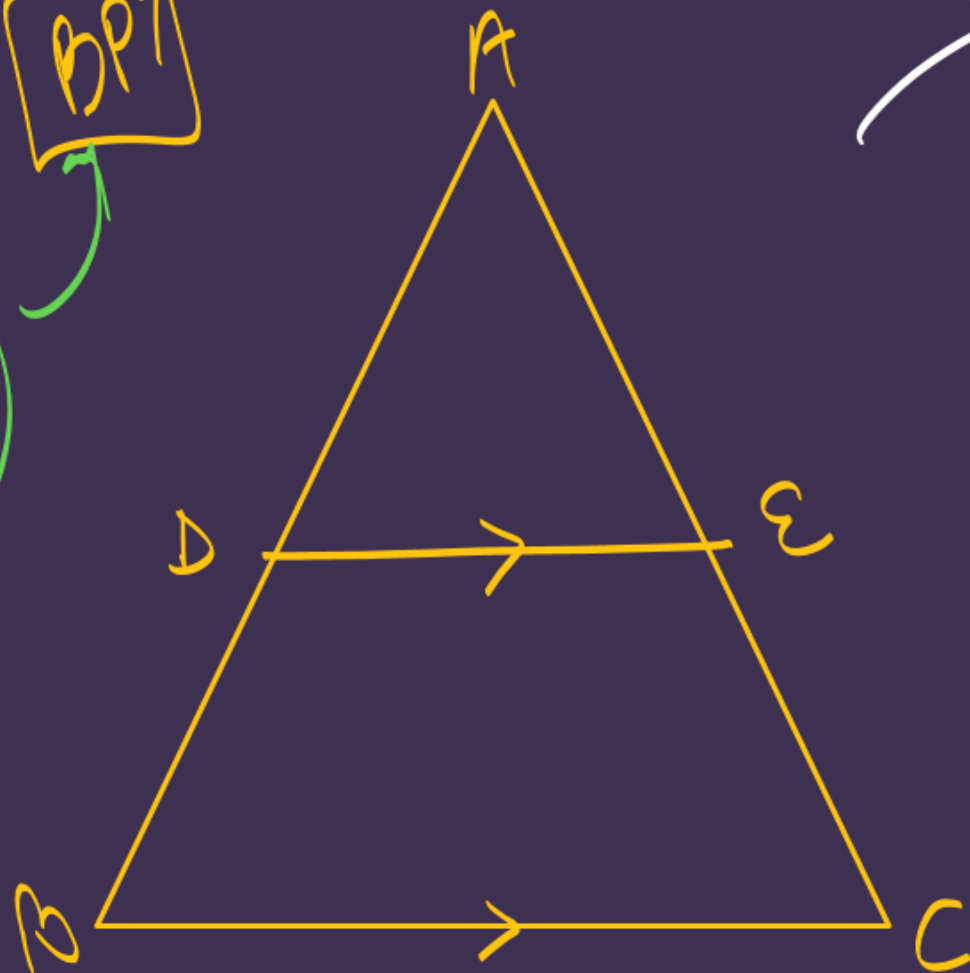
TRIANGLES

Lecture #2

ATK²:-

BPT

|| given



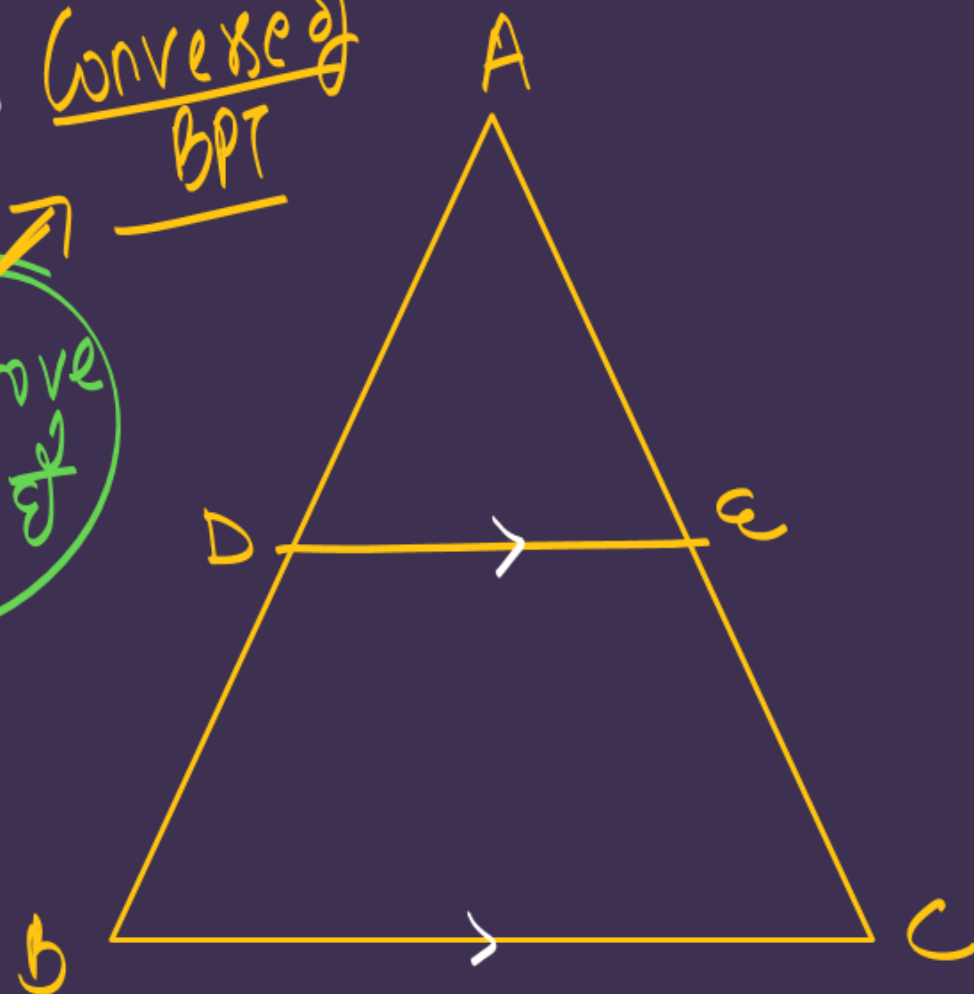
if $DE \parallel BC$

$$\frac{AD}{DB} = \frac{AE}{EC}$$

Converse

Converse of
BPT

|| to prove
that



if $\frac{AD}{DB} = \frac{AE}{EC}$

then, $DE \parallel BC$

#LP: X and Y are points on the sides AB and AC respectively of a triangle ABC such that $\frac{AX}{AB} = \frac{1}{4}$, $AY = 2$ cm and $YC = 6$ cm. Find whether $XY \parallel BC$ or not. (CBSE 2015)

given $\rightarrow \frac{AX}{AB} = \frac{1}{4} \checkmark$

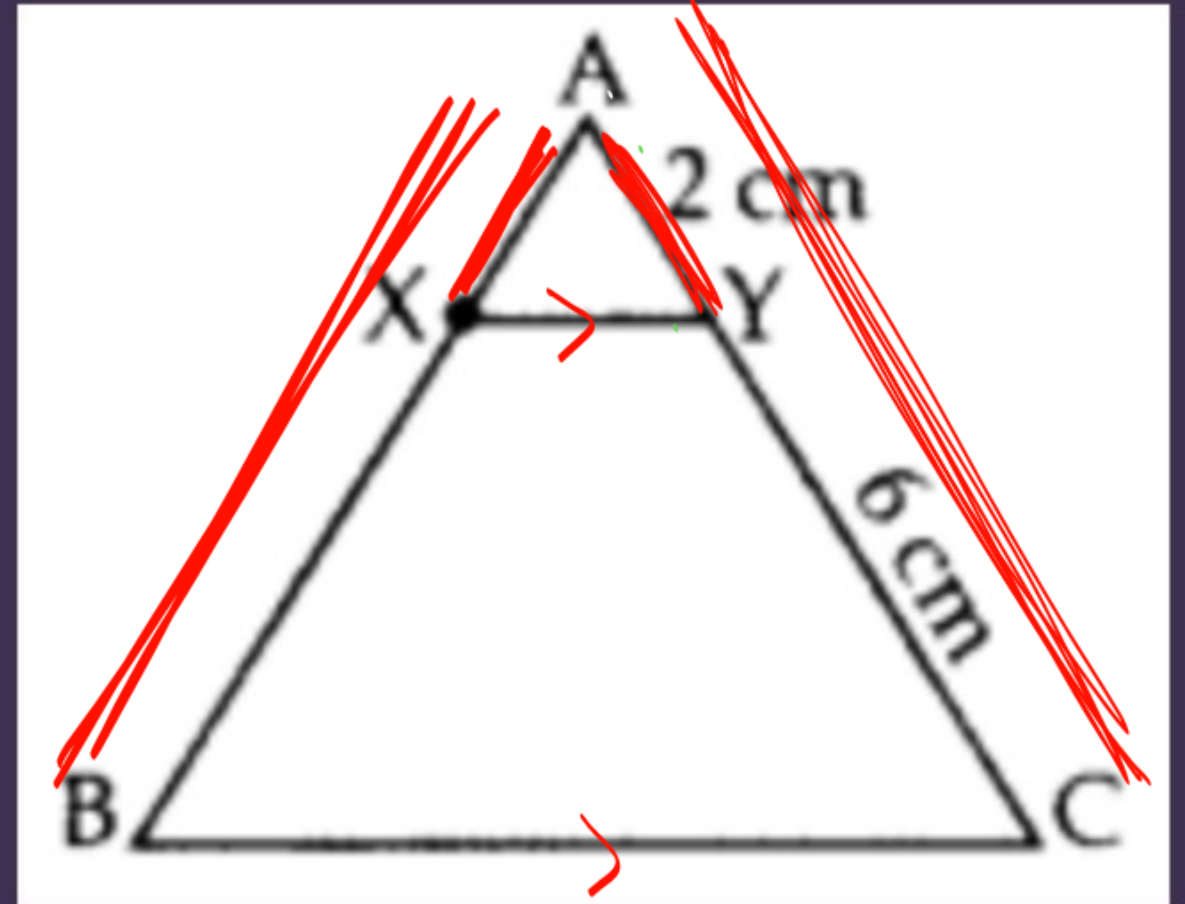
To check

$XY \parallel BC$??

$\frac{AY}{AC} \Rightarrow \frac{2}{2+6} \Rightarrow \frac{2}{8} = \frac{1}{4} \checkmark$

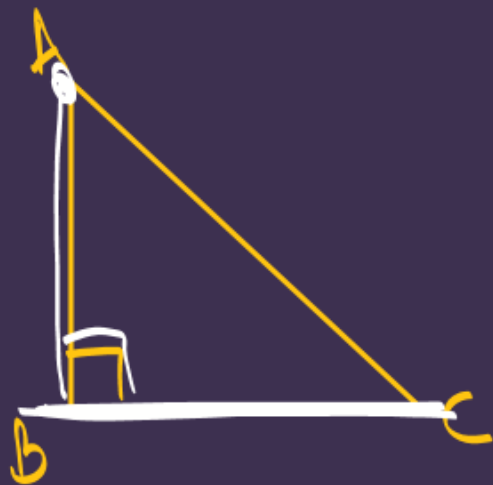
we can see $\Rightarrow \frac{AX}{AB} = \frac{1}{4}$ & $\frac{AY}{AC} = \frac{1}{4}$

by converse of BPT $\Rightarrow \boxed{XY \parallel BC}$



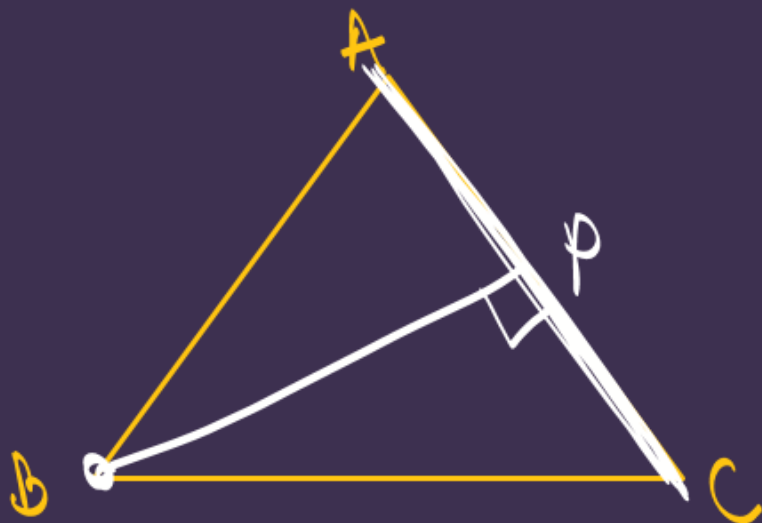
#100

$$\text{Ar. } \triangle = \frac{1}{2} \times b \times h \quad \leftarrow \text{height}$$



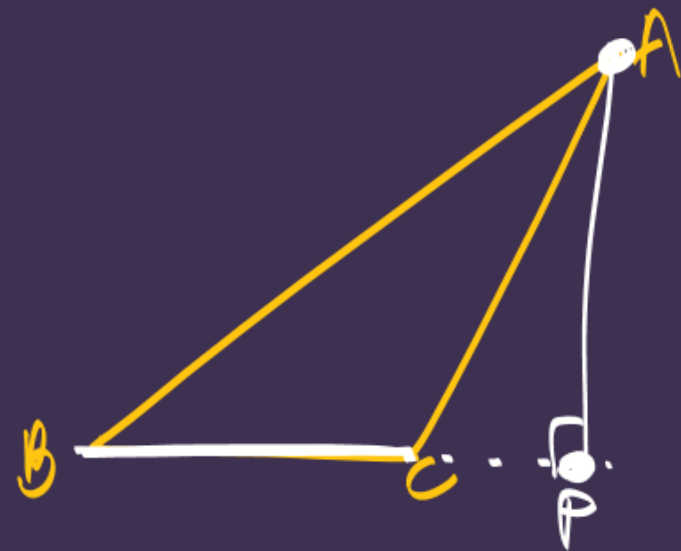
$$A = \frac{1}{2} \times b \times h$$

$$A = \frac{1}{2} \times BC \times AB$$



$$A = \frac{1}{2} \times b \times h$$

$$A = \frac{1}{2} \times AC \times BP$$

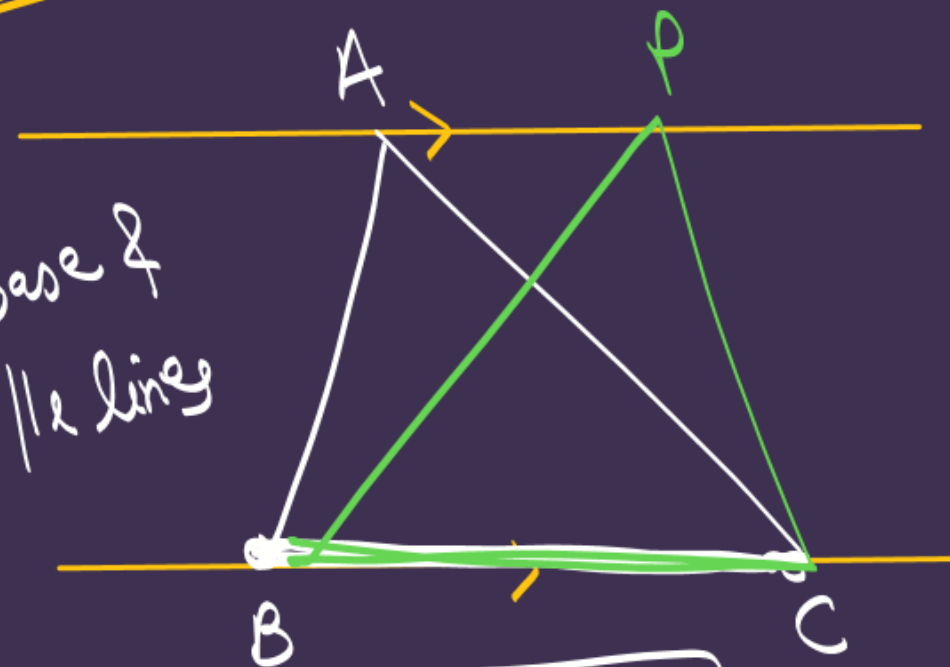


$$A = \frac{1}{2} \times b \times h$$

$$A = \frac{1}{2} \times BC \times AP$$

Imp theorem

two $\triangle$ s
having same base &
b/w some $\parallel$ lines



$$\text{ar}(\triangle ABC) = \text{ar}(\triangle PBC)$$

#BASIC PROPORTIONALITY THEOREM:

If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, then the other two sides are divided in the same ratio.

Given:- $DE \parallel BC$

To prove:- $\frac{AD}{DB} = \frac{AE}{EC}$

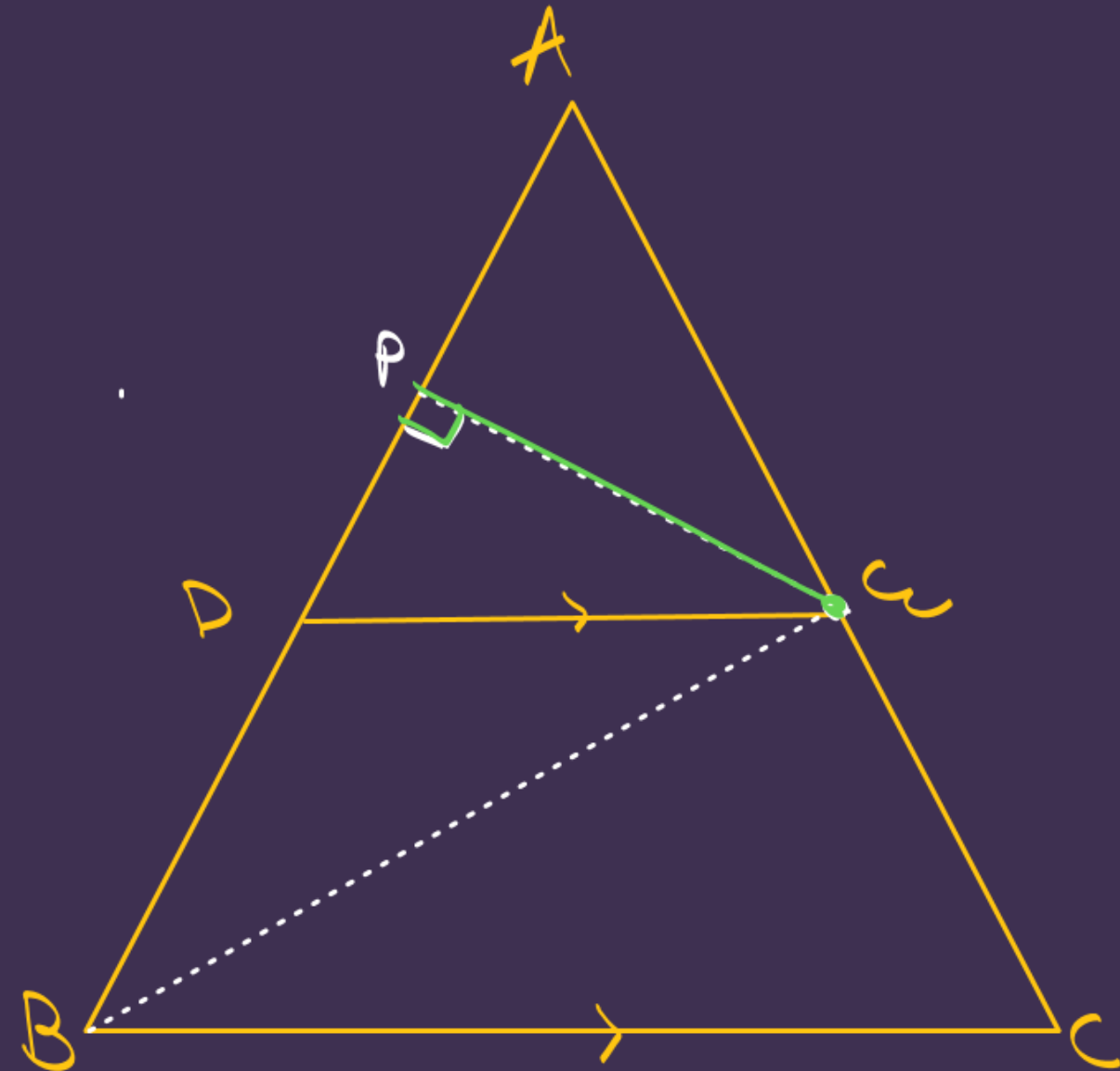
Construction:- Join BE , $EP \perp AB$

Proof:-

$$\text{ar}(\triangle ADE) = \frac{1}{2} \times AD \times EP$$

$$\text{ar}(\triangle EDB) = \frac{1}{2} \times BD \times EP$$

$$\Rightarrow \boxed{\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle EDB)} = \frac{AD}{BD}} \quad \text{--- (I)}$$



#BASIC PROPORTIONALITY THEOREM:

If a line is drawn parallel to one side of a triangle intersecting the other two sides in distinct points, then the other two sides are divided in the same ratio.

Given:- $DE \parallel BC$

To prove:- $\frac{AD}{DB} = \frac{AE}{EC}$

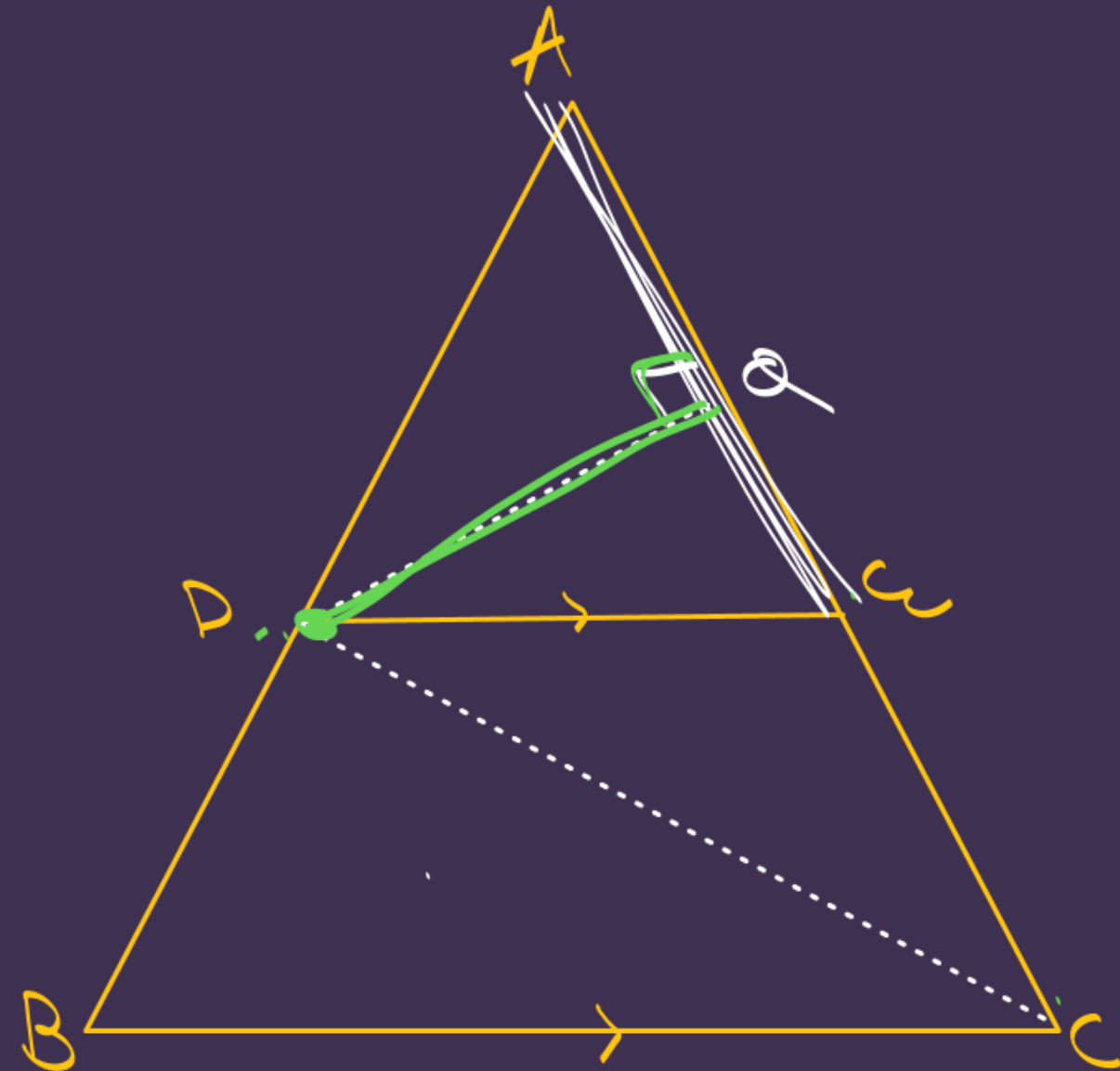
Construction:- Join BE, CD

Proof $\rightarrow$

$$\text{ar}(\triangle ADE) = \frac{1}{2} \times AE \times DQ$$

$$\text{ar}(\triangle DCE) = \frac{1}{2} \times CE \times DQ$$

$$\rightarrow \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DCE)} = \frac{AE}{CE} \quad \text{--- (I)}$$



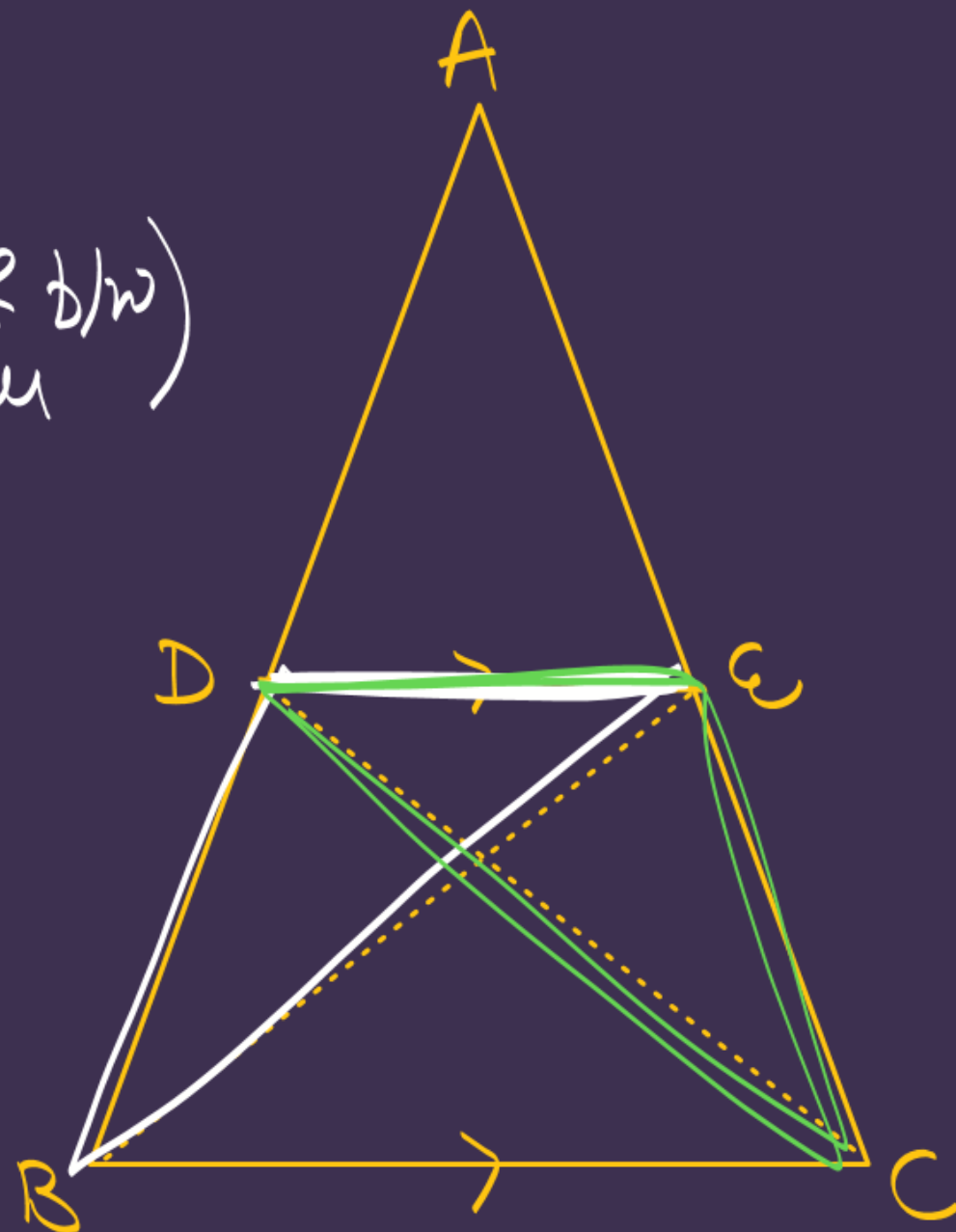
$$\textcircled{I} \rightarrow \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle EDB)} = \frac{AD}{BD}$$

$$\textcircled{II} \Rightarrow \frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DCE)} = \frac{AE}{CE}$$

$$\boxed{\text{ar}(\triangle EDB) = \text{ar}(\triangle DCE)} \quad \left(\begin{array}{l} \because \text{same base \& h/w} \\ \text{same || lines} \end{array} \right)$$

$\textcircled{I} \& \textcircled{II} \rightarrow \text{LHS} \begin{cases} \text{Num} \rightarrow \text{same} \\ \text{Den} \rightarrow \text{equal} \end{cases}$

$$\therefore \frac{AD}{BD} = \frac{AE}{CE}$$



#CONVERSE OF BPT:

If a line divides any two sides of a triangle in the same ratio, then the line must be parallel to the third side.

Given $\Rightarrow \frac{AD}{DB} = \frac{AE}{EC}$

To prove $\Rightarrow DE \parallel BC$

Proof: Let $DG \parallel BC$

BPT $\Rightarrow \frac{AD}{DB} = \frac{AG}{GC}$

but given $\frac{AD}{DB} = \frac{AE}{EC}$

$1 + \frac{AG}{GC} = \frac{AE}{EC} + 1$

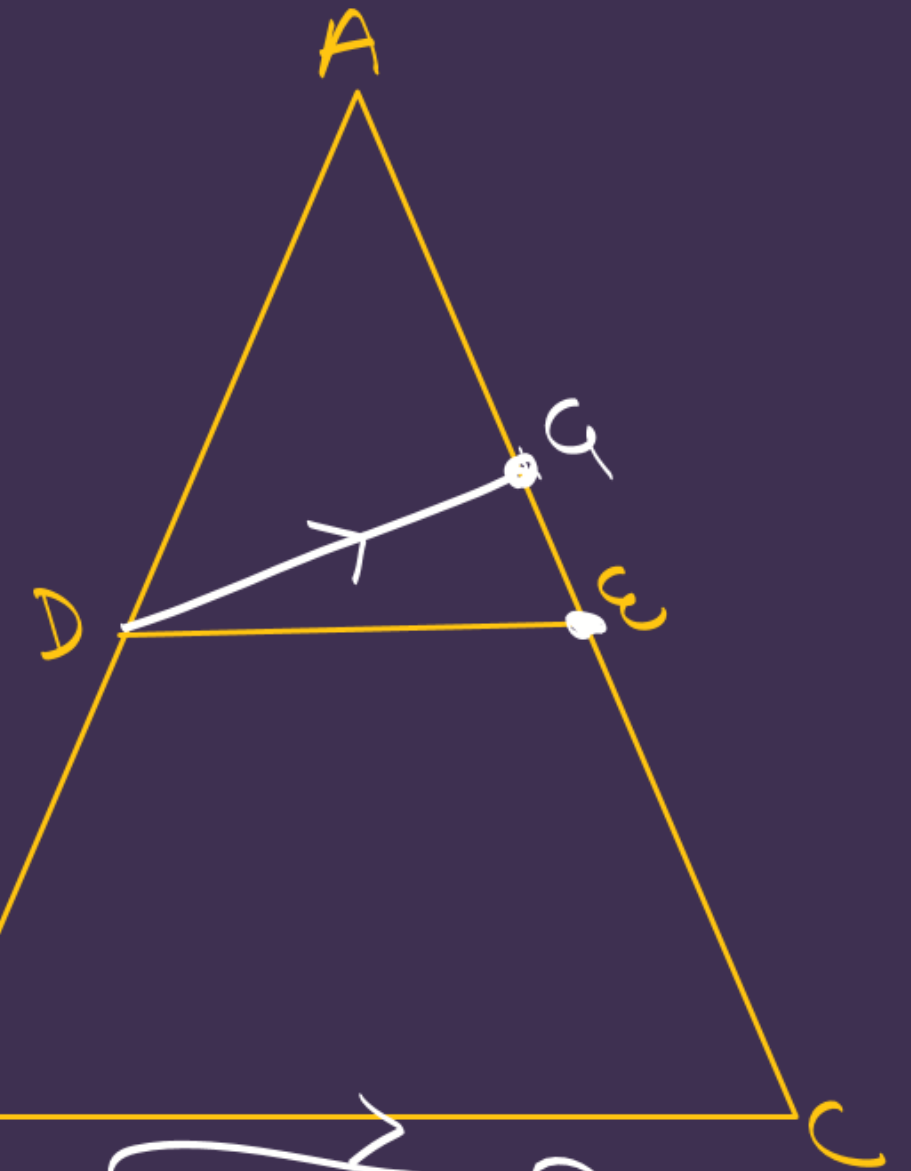
add ① both sides

$\frac{GC + AG}{GC} = \frac{AE + EC}{EC}$

$\frac{\cancel{AG}}{GC} = \frac{\cancel{AE}}{EC}$

$\Rightarrow \boxed{EC = GC}$

i.e. $\textcircled{G}$ coincide with $\textcircled{E}$



i.e. $\boxed{DE \parallel BC}$ ~~HP~~

#LP: Let ABC be a triangle and D and E be two points on side AB such that $AD = BE$. If $DP \parallel BC$ and $EQ \parallel AC$, then prove that $PQ \parallel AB$.

In $(\triangle ABC)$ BPT

$$\frac{AD}{DB} = \frac{AP}{PC}$$

$$\Rightarrow \frac{BE}{DE + BE} = \frac{AP}{PC}$$

$$\Rightarrow \frac{BE}{DE + AD} = \frac{AP}{PC}$$

$$\Rightarrow \frac{BE}{AE} = \frac{AP}{PC} \quad (1)$$

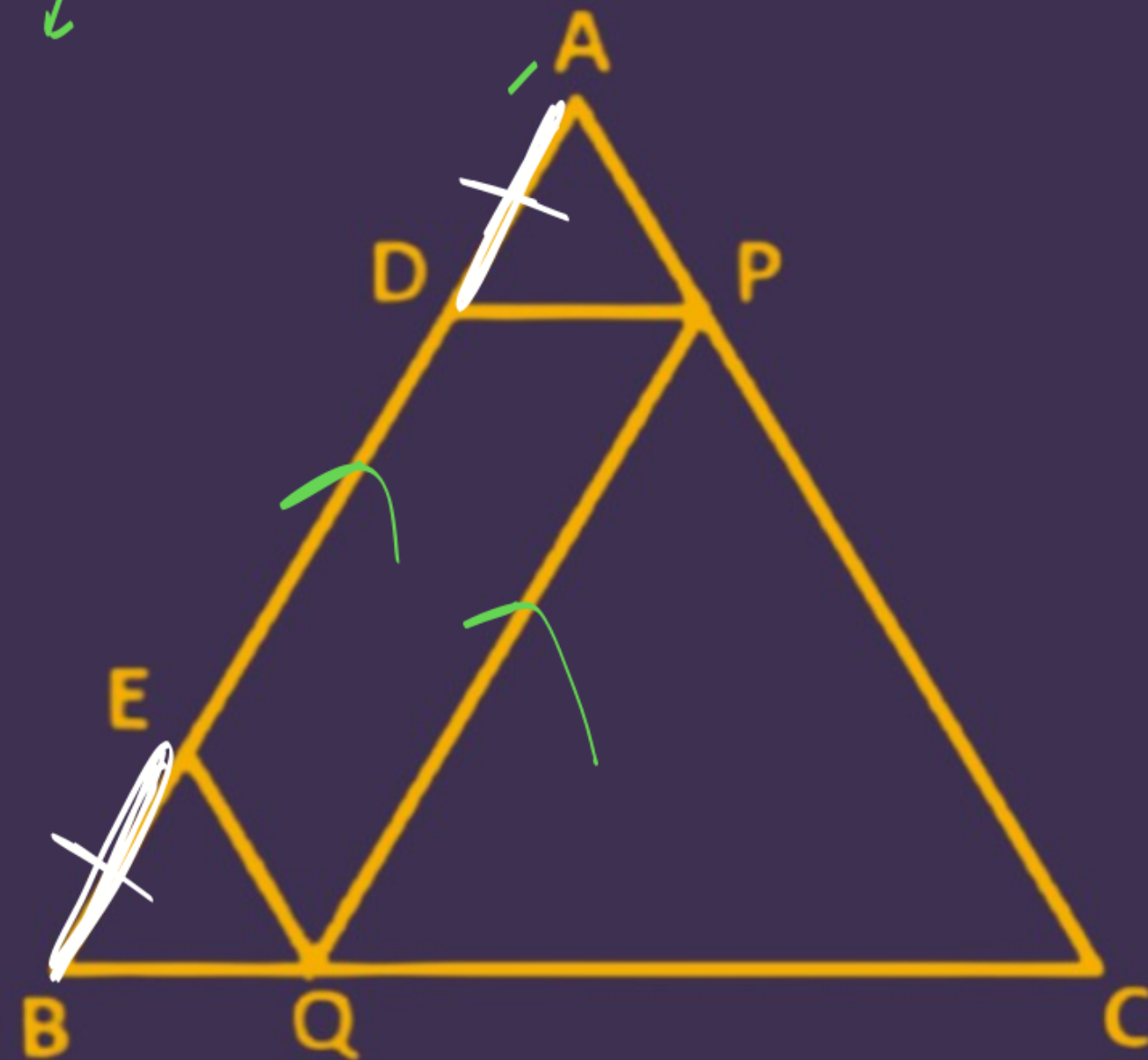
$(\triangle ABC)$ BPT

$$\frac{BQ}{QC} = \frac{BE}{AE} \quad (2)$$

from (1) & (2)

$$\frac{AP}{PC} = \frac{BQ}{QC}$$

by conv. of BPT $\Rightarrow PQ \parallel AB$ (RP)



#LP: $DE \parallel AC$ and $DC \parallel AP$ prove that $\underline{BE/EC = BC/CP}$

[CBSE]

($\triangle ABC$) BPT

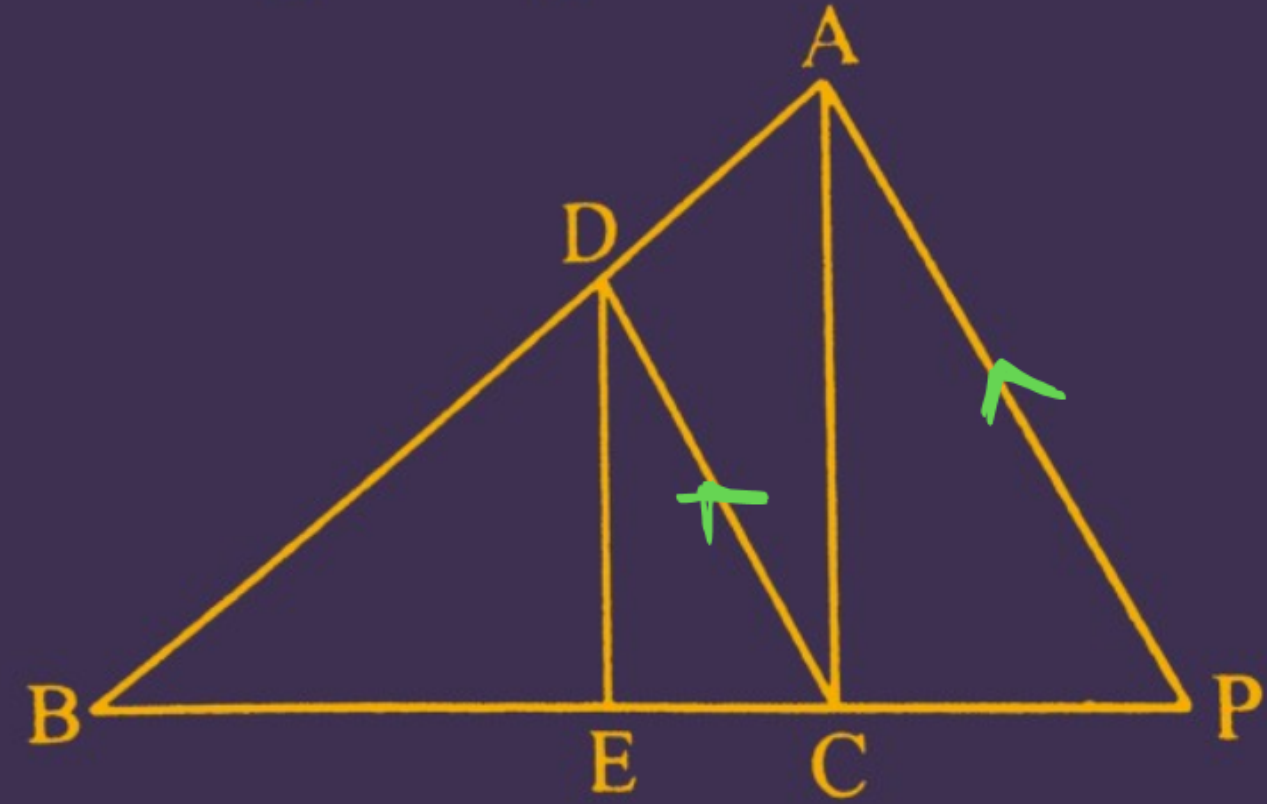
$$\frac{BE}{EC} = \frac{BD}{DA} \quad \text{--- (I)}$$

($\triangle ABP$) BPT

$$\frac{BC}{CP} = \frac{BD}{DA} \quad \text{--- (II)}$$

from (I) & (II)

$$\frac{BE}{EC} = \frac{BC}{CP} \quad \text{Q.P.D}$$



#LP: ABCD is a trapezium in which $AB \parallel DC$ and its diagonals intersect each other at the point O. Show that $AO/CO = BO/DO$

Given: $AB \parallel DC$

To prove: $\frac{AO}{CO} = \frac{BO}{DO}$ ✓

Construction: Draw $OP \parallel AB$

Proof: In $\triangle DAB$, $OP \parallel AB$

by BPT

$$\frac{DP}{PA} = \frac{DO}{OB} \quad (1)$$

$OP \parallel CD$ also
($\triangle ACD$)

BPT $\Rightarrow \frac{PA}{DP} = \frac{AO}{CO}$

reciprocal $\frac{DP}{PA} = \frac{CO}{AO} \quad (2)$

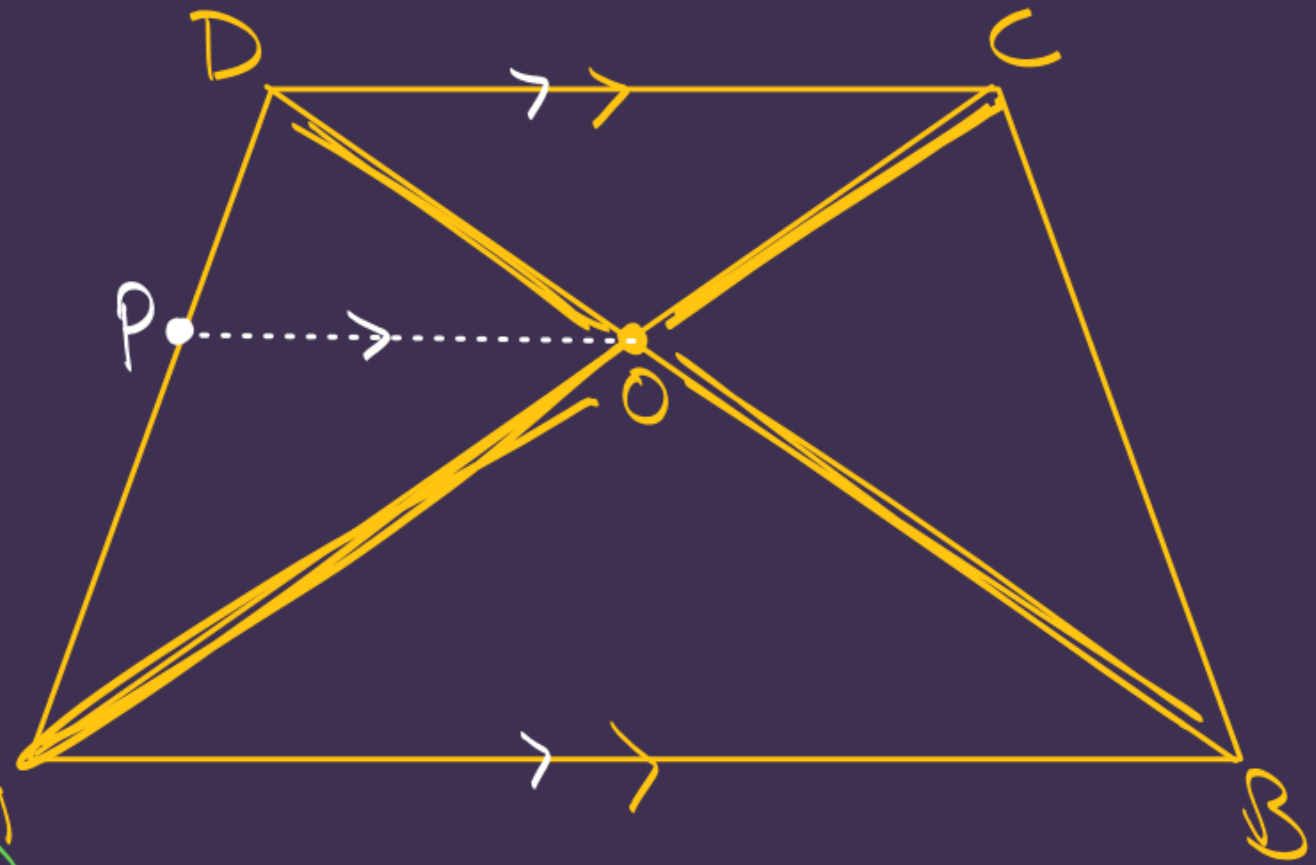
From (1) & (2)

$$\frac{DO}{OB} = \frac{CO}{AO}$$

reciprocal

$$\frac{OB}{DO} = \frac{AO}{CO}$$

Q.E.D.



#LP: The diagonals of a quadrilateral ABCD intersect each other at the point O such that $\frac{AO}{CO} = \frac{BO}{DO}$. Show that ABCD is a trapezium

Given $\Rightarrow \frac{AO}{CO} = \frac{BO}{DO}$ — (I)

To prove: ABCD is Trap. $\therefore \underline{AB \parallel CD}$

Construction: Draw $OP \parallel AB$ — (II)

from (I) & (II)

Proof: $\triangle DAB, OP \parallel AB$

BPT

$$\frac{DP}{PA} = \frac{DO}{BO}$$

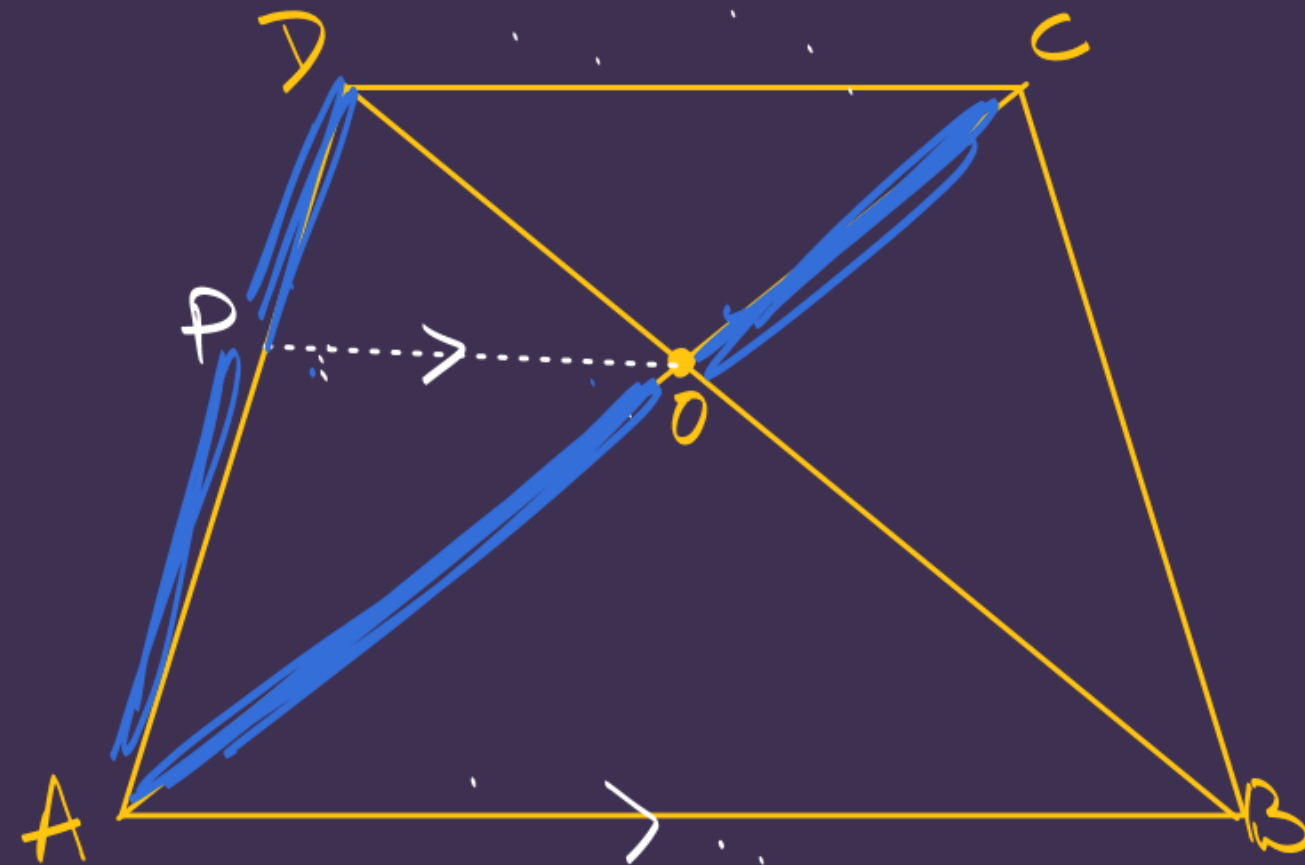
reciprocal

$$\frac{PA}{OP} = \frac{BO}{DO}$$

$$\frac{AO}{CO} = \frac{PA}{DP}$$

$\rightarrow$ by conv. of BPT

$OP \parallel CD$ — (III)



from (II) & (III)

$\underline{AB \parallel CD}$ — (IV)

ABCD is
Trap.

#LP: Let X be any point on the side BC of a triangle ABC . If $\underline{XM}$, $\underline{XN}$ are drawn parallel to $\underline{BA}$ and $\underline{CA}$, $\underline{BA}$ in M , N respectively; MN meets BC produced in T , prove that $\underline{TX^2 = TB \times TC}$. ← To prove

Given *

$XM \parallel BA$

→ $XM \parallel BN$

BPT

$$\frac{TB}{TX} = \frac{TN}{TM} \quad \text{--- (I)}$$

$XN \parallel CA$

↳

$XN \parallel CM$

also

BPT

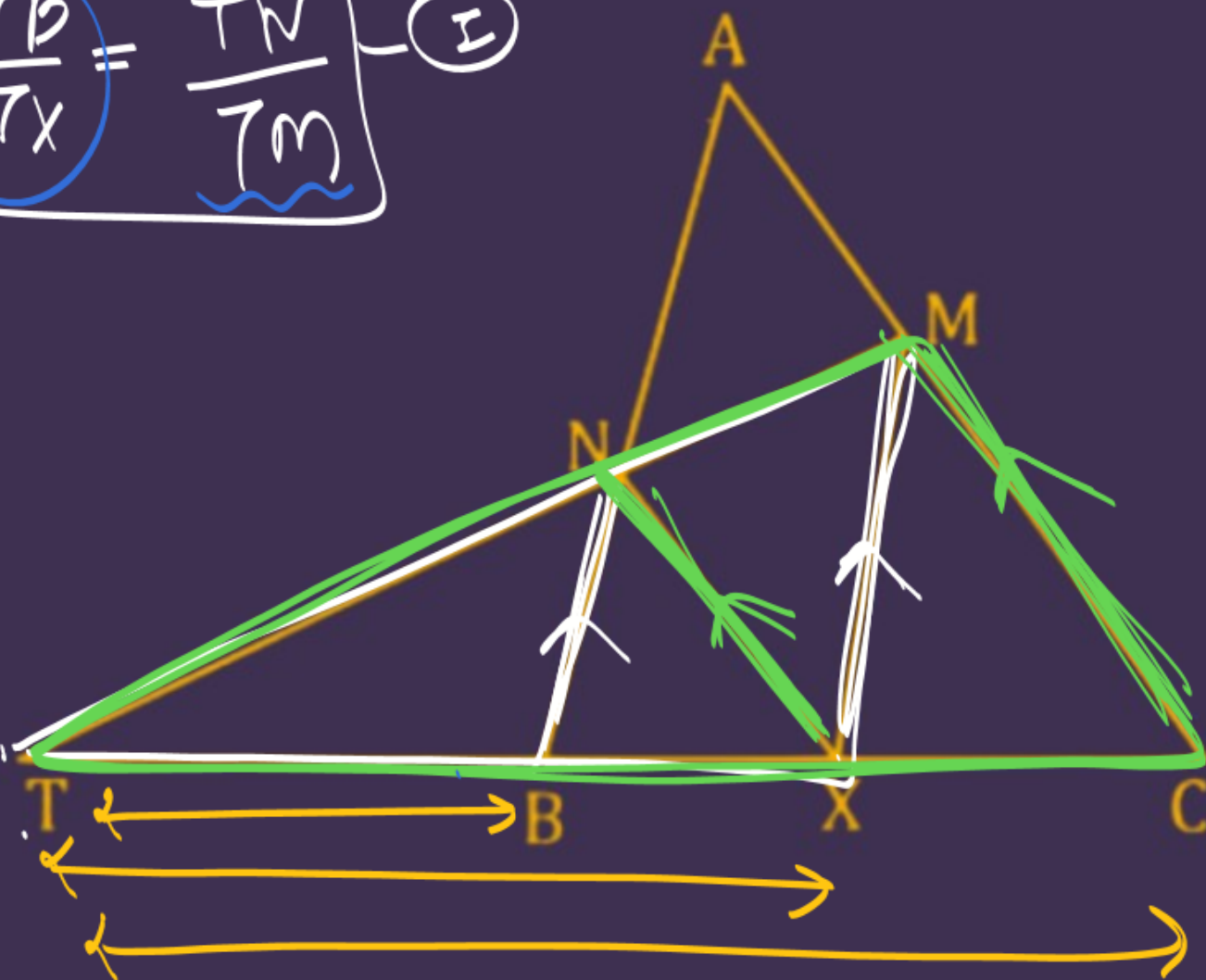
$$\frac{TX}{TC} = \frac{TN}{TM} \quad \text{--- (II)}$$

from (I) & (II)

$$\frac{TB}{TX} = \frac{TX}{TC}$$

$$TB \times TC = (TX)(TX)$$

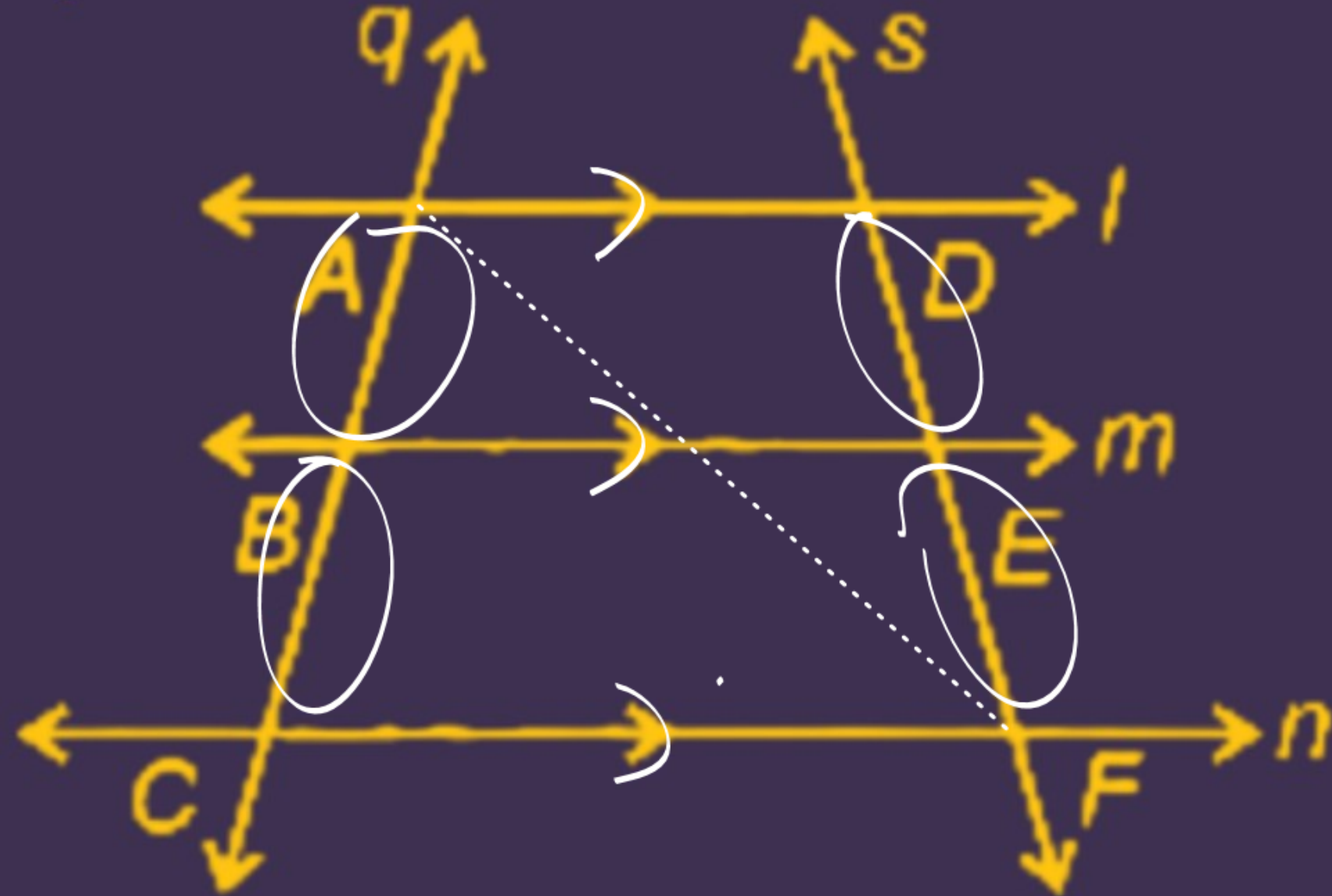
$$TB \times TC = TX^2 \quad \text{--- (Proved)}$$





#LP: State basic proportionality theorem. Use it prove the following: If three parallel lines l, m, n are intersected by transversals q and s as shown in the figure, then $AB/BC = DE/EF$

Join AF.



MPT

#LP: Prove that the line joining the mid-points of any two sides of a triangle is parallel to the third side. (Recall that you have done it in Class IX)

Given $AP = PB$ & $AQ = QC$

To prove:-

$PQ \parallel BC$ $\rightarrow$  Conv. of BPT.

Proof :-

Given $\Rightarrow$

$$AP = PB$$

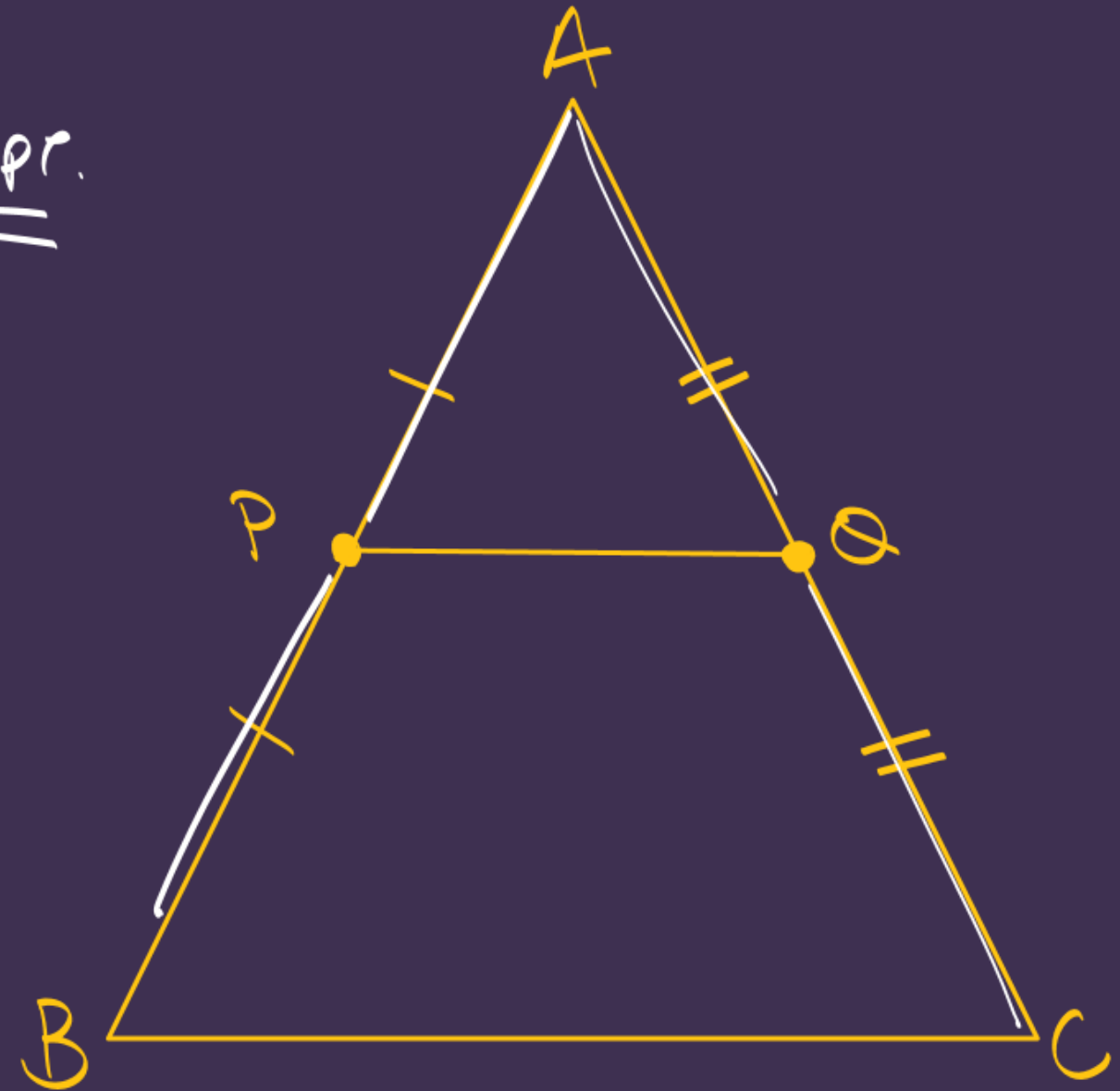
$$AQ = QC$$

$$\frac{AP}{PB} = 1$$

$$\frac{AQ}{QC} = 1$$

$$\frac{AP}{PB} = \frac{AQ}{QC}$$

by Conv. of BPT $\Rightarrow$ $PQ \parallel BC$ ~~PP~~



Converse of MPT

#LP: Prove that a line drawn through the mid-point of one side of a triangle parallel to another side bisects the third side. (Recall that you have proved it in Class IX).

Given:-

P is MP of AB $\Rightarrow$ $\boxed{AP = PB}$
PQ || BC ✓

To prove:

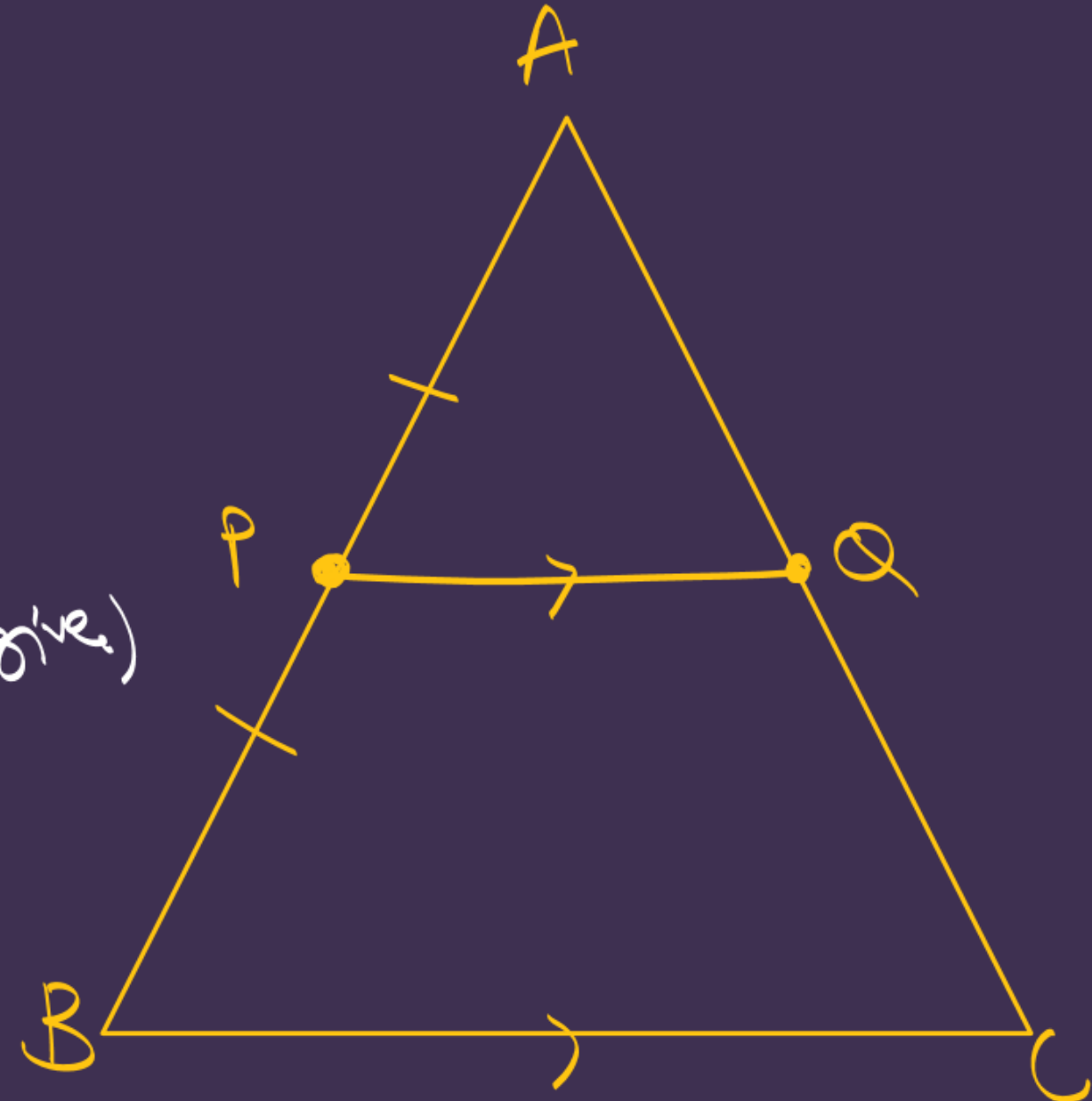
$\boxed{AQ = QC}$ ✓

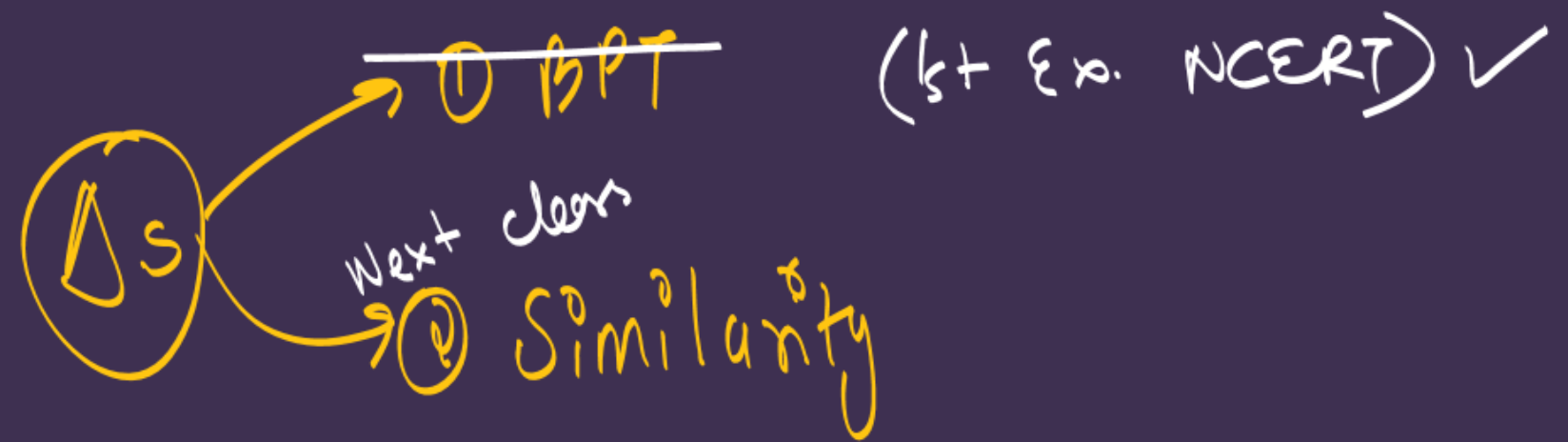
Proof:-

Since PQ || BC

by BPT $\frac{AP}{PB} = \frac{AQ}{QC}$ ($\because AP = PB$ given)

$\boxed{QC = AQ}$ ✓ HP





Thank you Goodies !!

